

Chapter 3 Notes: Systems with Feedback

3.0 The General Feedback Block Diagram

A plant is simplified via block-diagram algebra into a single transfer function, then a controller is added with negative feedback so the overall system looks like:

$R(s) \rightarrow [\Sigma] \rightarrow \text{Controller} \rightarrow U(s) \rightarrow \text{Plant} \rightarrow Y(s)$, with the output fed back through a **Sensor** to the summing junction (subtracted from $R(s)$).

Three internal signals

1. **$E(s)$ — the error signal** When feedback is negative, it's the difference between the reference and the measured output:

$$E(s) = R(s) - Y_m(s)$$

2. **$U(s)$ — the control effort** The controller output that drives the plant:

$$U(s) = \text{Controller} \cdot E(s)$$

3. **$Y_m(s)$ — the measured output signal**, used for feedback:

$$Y_m(s) = \text{Sensor} \cdot Y(s)$$

(Often the difference between the true output and the measured output is insignificant for design purposes.)

Two external signals

1. **$R(s)$ — the reference input signal**
2. **$Y(s)$ — the controlled system output signal:**

$$Y(s) = \text{Plant} \cdot U(s)$$

3.1 Simplified Block Diagram and Closed-Loop vs. Open-Loop Forms

The wordy diagram above simplifies to the standard **unity-ish negative feedback loop**:

$R(s) \rightarrow [\Sigma, -] \rightarrow G_c(s) \rightarrow G(s) \rightarrow Y(s)$, with $H(s)$ feeding back from $Y(s)$ to the summing junction.

- $G_c(s)$ = controller
- $G(s)$ = plant
- The feedback loop (output routed back to the summer) makes this a **closed-loop model**.

Reducing to a single block

$$R(s) \longrightarrow \boxed{\frac{G_c G(s)}{1 + G_c G H(s)}} \longrightarrow Y(s)$$

When this ratio is simplified into a single ratio of polynomials, it's called the **open-loop representation**:

$$R(s) \longrightarrow \boxed{\frac{B(s)}{A(s)}} \longrightarrow Y(s)$$

Key distinctions

	Closed-loop form	Open-loop form
Structure	Retains forward path $G_c G(s)$ and loop-gain $L(s) = G_c G H(s)$ separately	Collapsed into a single ratio of polynomials $B(s)/A(s)$
Loop-gain	Directly observable	Lost — cannot recover G, H uniquely
Use case	Analyzing/designing loop-gain behavior	Simplified analysis once design is done

- Both forms are mathematically valid and lossless — you can transform between them.
- $A(s)$ (the open-loop denominator) has a special name: the **characteristic polynomial**.

3.2 Closed-Loop → Open-Loop (Example 1)

Given:

$$G_c(s) = \frac{3}{s+3}, \quad G(s) = \frac{10}{s+1}, \quad H(s) = \frac{2}{s+2}$$

Step 1 — Forward path:

$$G_c G(s) = \frac{3 \cdot 10}{(s+3)(s+1)} = \frac{30}{s^2 + 4s + 3}$$

Step 2 — Loop-gain:

$$G_c G H(s) = \frac{3 \cdot 10 \cdot 2}{(s+3)(s+1)(s+2)} = \frac{60}{s^3 + 6s^2 + 11s + 6}$$

Step 3 — Closed-loop form using $T(s) = \frac{G_c G(s)}{1 + G_c G H(s)}$:

$$T(s) = \frac{\frac{30}{s^2+4s+3}}{1 + \frac{60}{s^3+6s^2+11s+6}}$$

Step 4 — Simplify to open-loop (polynomial) form:

$$T(s) = \frac{30(s+2)}{s^3 + 6s^2 + 11s + 66} = \frac{B(s)}{A(s)} = \frac{30s + 60}{s^3 + 6s^2 + 11s + 66}$$

Takeaway: the closed-loop expression preserves $G_c G(s)$ and $L(s) = G_c G H(s) = \frac{60}{s^3 + 6s^2 + 11s + 66}$ as separate pieces; the open-loop (simplified) form loses that separation entirely.

3.3 Open-Loop → Closed-Loop: The Continued Fraction Model

Going the *other* direction (given only a single open-loop $T(s)$, find a plausible $G(s)/H(s)$ pair) uses the **continued fraction model**.

General procedure:

1. Divide numerator and denominator of $T(s)$ by the numerator, so

$$T(s) = \frac{1}{(\text{denominator})/(\text{numerator})}.$$

2. Split the resulting denominator fraction into **two terms** — make one term as simple as possible (ideally a constant).
3. Match this to the standard closed-loop form:

$$T(s) = \frac{1}{\frac{1}{G(s)} + H(s)}$$

4. **Choose $H(s)$ to be the simplest term** (this becomes the sensor/feedback transfer function); the other term becomes $\frac{1}{G(s)}$.

Example 2 (strictly proper denominator)

$$T(s) = \frac{s + 4}{s^2 + s + 9}$$

Rewrite: $T(s) = \frac{1}{\frac{s^2+s+9}{s+4}}$, split as $\frac{1}{\frac{s^2+s}{s+4} + \frac{9}{s+4}}$

Choosing the simpler term as $H(s)$:

$$H(s) = \frac{9}{s + 4}, \quad G(s) = \frac{s + 4}{s^2 + s}$$

Loop-gain: $L(s) = GH(s) = \frac{9}{s^2 + s}$

Example 3 (simpler case)

$$T(s) = \frac{4}{s + 13}$$

Split: $T(s) = \frac{1}{\frac{s}{4} + \frac{13}{4}}$

$$H(s) = \frac{13}{4}, \quad G(s) = \frac{4}{s}, \quad L(s) = \frac{13}{s}$$

Example 4 — Biproper Case

$$T(s) = \frac{s^2 + 3s + 9}{s^2 - 2s - 7}$$

Here the numerator and denominator have the **same order** — this is called **biproper**. You cannot split it directly; **long division first**:

$$T(s) = 1 + \frac{5s + 16}{s^2 - 2s - 7} = 1 + \hat{T}(s)$$

Then apply the continued fraction model to the **remainder** $\hat{T}(s)$ only:

$$\hat{T}(s) = \frac{1}{\frac{s^2-2s}{5s+16} - \frac{7}{5s+16}}$$

$$H(s) = -\frac{7}{5s+16}, \quad G(s) = \frac{5s+16}{s^2-2s}, \quad L(s) = -\frac{7}{s^2-2s}$$

Block diagram note: because $T(s) = 1 + \hat{T}(s)$, the "1" becomes a **direct feed-forward path** from input straight to output (bypassing the loop entirely), summed with the output of the feedback loop for $\hat{T}(s)$. Also — since $H(s)$ here is *negative*, the feedback through $H(s)$ ends up **positive** overall (a negative times a negative sign in the loop cancels out).

Example 5 — Improper Case

$$T(s) = \frac{s^3 + s^2 + 9s + 9}{s^2 + 4s + 2}$$

Here the **denominator's order is lower than the numerator's** — this is **improper**.

There is no way to construct a closed-loop model for an improper open-loop transfer function. (Physically, this would require pure differentiators with no bandwidth limit — not realizable.)

3.1 (cont'd) — Choosing H(s): Practical Guidance

When you have a choice of how to split the continued fraction, **always pick the simplest expression for $H(s)$** . Ideally:

- **Unity feedback** ($H(s) = 1$) is best, if the model allows it.
- Otherwise, choose the simplest possible model — usually a constant over a low-order polynomial.

Why simpler $H(s)$ is preferred:

1. **Minimizes feedback circuitry** — simpler feedback introduces less lag.
2. **Avoids differentiation** — differentiators amplify high-frequency noise.

Differentiation-and-Noise Example

Given $\frac{Y(s)}{R(s)} = \frac{1}{s + 13}$, two options exist for the closed-loop split:

Option 1: $G(s) = \frac{1}{s}$ (an integrator in the forward path), $H(s) = 13$

Option 2: $G(s) = \frac{1}{13}$, $H(s) = s$ (a **differentiator** in the feedback path)

Mathematically these are equivalent, **but Option 1 is far better in practice.**

Why — the noise argument:

- ***Option 1 (integrator):*** if the error signal is corrupted by high-frequency noise,

$$e(t) = \sin(t) + 0.01 \sin(10^{10}t)$$

then integrating gives

$$y(t) = -\cos(t) - \frac{0.01}{10^{10}} \cos(10^{10}t) \approx \text{signal}$$

The noise term's amplitude is ****divided**** by the huge frequency — integration ****attenuates**** high-frequency noise.

- ***Option 2 (differentiator):*** if instead the ***output*** signal is corrupted,

$$y(t) = \sin(t) + 0.01 \sin(10^{10}t)$$

differentiating gives

$$Y_m(t) = \cos(t) + (0.01 \cdot 10^{10}) \cos(10^{10}t) \approx \text{noise}$$

The noise term's amplitude gets ****multiplied**** by the huge frequency — differentiation ****amplifies**** high-frequency noise catastrophically.

Conclusion: integrators suppress high-frequency noise; differentiators amplify it. Since high-frequency noise is common in real electronics (switching power supplies run at 200 kHz–2 MHz, plus oscillators/timers), **always prefer the option that avoids differentiation in the loop.**

3.1.1 Minimal Realization (Pole-Zero Cancellation)

Block-diagram manipulation can produce an open-loop transfer function where a pole and a zero coincide (cancel). **Be careful** — only some cancellations are valid.

Rule: only *stable* poles may be canceled. Unstable poles must never be canceled.

Stable Cancellation (allowed)

$$T(s) = \frac{s^2 + 7s + 6}{s^3 + 18s^2 + 104s + 192} = \frac{(s + 1)(s + 6)}{(s + 8)(s + 6)(s + 4)}$$

The $(s + 6)$ factor is a **stable** pole (negative real part) that cancels with a matching zero — safe to simplify:

$$T(s) = \frac{s + 6}{(s + 8)(s + 4)} = \frac{s + 6}{s^2 + 12s + 32}$$

Unstable Realization (cancellation NOT allowed)

$$T(s) = \frac{s^2 - 5s - 6}{s^3 + 6s^2 - 40s - 192} = \frac{(s + 1)(s - 6)}{(s + 8)(s - 6)(s + 4)}$$

Here $(s - 6)$ is an **unstable** pole (positive real part, $s = +6$). Even though it algebraically cancels with a zero, doing so **hides an unstable mode** of the real system — in reality it can never cancel perfectly (component

tolerances, rounding, etc.). To reflect this, the pole is perturbed slightly instead of canceled outright:

$$T(s) = \frac{(s + 1)(s - 6 + \varepsilon)}{(s + 8)(s - 6 + \varepsilon)(s + 4)}$$

Why this matters: in a physical system, an "exact" unstable pole-zero cancellation is never truly exact. Any tiny mismatch (ε) leaves a lightly-damped or growing mode in the system that can blow up in practice — even though the idealized math says it cancels. This is why unstable pole-zero cancellations must never be assumed away in analysis or design.

Quick-Reference Summary

Concept	Key takeaway
Closed-loop form	$T(s) = \frac{G_c G}{1 + G_c G H}$ — keeps forward path & loop-gain visible
Open-loop form	$T(s) = \frac{B(s)}{A(s)}$ — single polynomial ratio; $A(s)$ = characteristic polynomial
Continued fraction model	Used to go open-loop → closed-loop ; split denominator into two terms, simplest one → $H(s)$
Biproper $T(s)$	Numerator & denominator same order → long-divide first, remainder becomes feed-forward "+1" path
Improper $T(s)$	Numerator order > denominator order → no closed-loop model exists
Choosing $H(s)$	Prefer unity or simplest possible — minimizes lag, avoids differentiation
Integrator vs. differentiator	Integrators attenuate high-freq noise; differentiators amplify it — avoid differentiators in feedback
Pole-zero cancellation	Only cancel stable poles; never cancel unstable poles (not physically realizable)